Spotify Recommendations

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Approach:

In calculating similarity scores between our sampled songs, we initially took just the euclidean distance between songs' audio features as vectors. However, after perceiving these songs to be quite different to our idea of The Strokes' music, we wondered if there was a better way to approach the similarity score. We looked into research on what makes us perceive songs as "similar", and created weights corresponding to features in the Spotify database, with ratios similar to what research had identified. Thus, our top 5 song recommendations are the lowest euclidean distance between songs' audio features, with unique feature weighting.

Similarity Metric:

Euclidean distance (with feature weights)

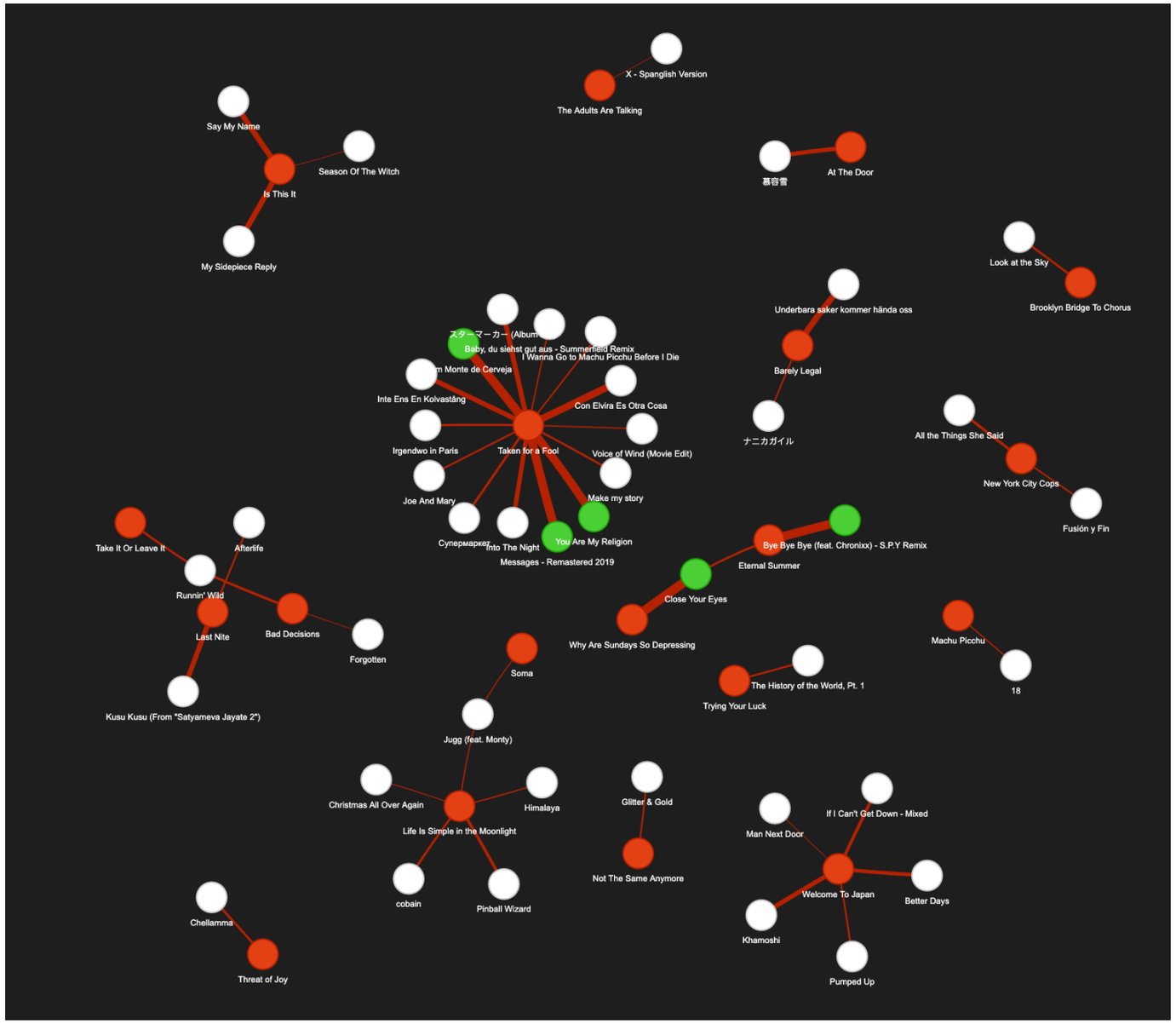
similarity = np.round(1 - (distances / distances.max()), 4)

Weights:

- danceability = 1.0 (baseline)
- energy = 2.5 (timbre - most important)
- loudness = 1.0 (baseline)
- speechiness = 0.5 (less relevant)
- acousticness = 2.5 (timbre - most important)
- instrumentalness = 1.0 (baseline)
- liveness = 0.5 (less relevant)
- valence = 1.5 (harmonic)
- tempo = 2.0 (rhythm)

 # Nodes = 1,500

 # Edges = 1,857



	Title	Artist	Album	Score
1	Close Your Eyes	Kim Petras	Happy Haunted Halloween Tunes	0.6053
2	Por um Monte de Cerveja	Celso Blues Boy	Por um Monte de Cerveja	0.5891
3	Bye Bye Bye (feat. Chronixx) - S.P.Y Remix	SHY FX;Chronixx;S.P.Y	Raggamuffin Reloaded	0.5796
4	Messages - Remastered 2019	Orchestral Manoeuvres In The Dark	Souvenir	0.5766
5	You Are My Religion	Firehouse	Good Acoustics	0.5527